

EES301: Global biogeochemical cycles and Climate change

Human modifications of the Earth system **Large-scale eutrophication and food production**

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EES301

Element	Juvenile flux ^a (1)	Chemical weathering (2)	Natural cycle ^b (3)	Biospheric recycling ratio ^c 3/(1+2)	Human mobilization ^d (4)	Human enhancement 4/(1+2)
B	0.02	0.19	8.8	42	0.58	2.8
C	30	210	107,000	446	8700	36.3
N	5	20 ^e	9200 ^f	368	221	8.8
P	~0	2	1000	500	25	12.5
S	10	70	450	5.6	130	1.6
Cl	2	260	120	0.46	170	0.65
Ca	120	500	2300	3.7	65	0.10
Fe	6	1.5	40	5.3	1.1 ^g	0.14
Cu	0.05	0.056	2.5	23.6	1.5 ^g	14.2
Hg	0.0005	0.0002	0.003	4.3	0.0023	3.3

Estimates of the global flux in the Biogeochemical cycles of certain elements illustrating the human impact (in 10¹² g/yr)

Source: Biogeochemistry: An Analysis of Global Change

Note: All data 10¹² g/yr.

^a Degassing from the Earth's crust and mantle; sum of volcanic emissions to the atmosphere (subaerial) and net hydrothermal (Elderfield and Schultz 1996) and for N, fixation by lightning (Chapter 12).

^b Annual biogeochemical cycle to/from the Earth's biota on land and in the oceans, in the absence of humans.

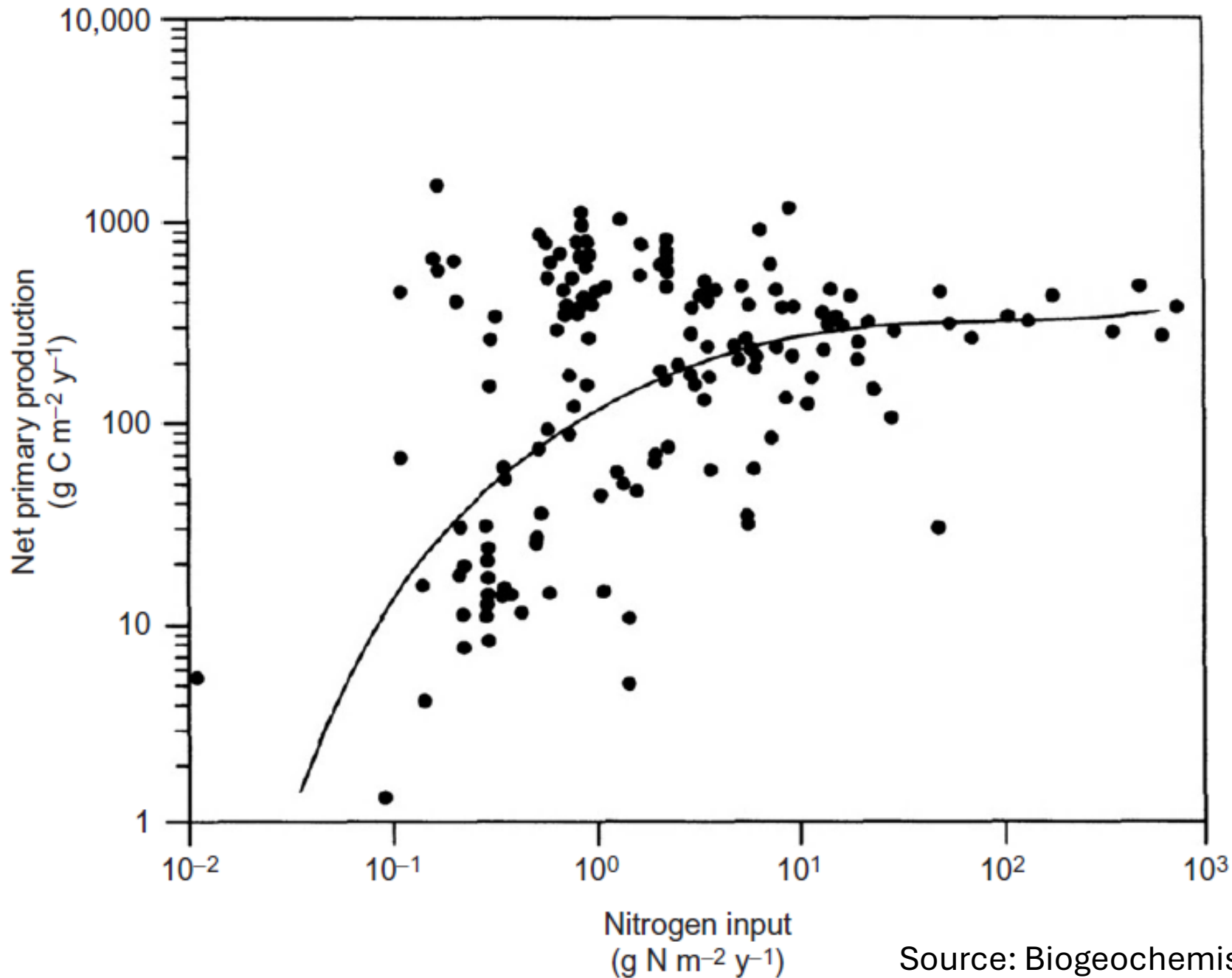
^c Following Volk (1998).

^d Direct and indirect mobilization by extraction and mining from the Earth's crust or (for N) industrial fixation. (Sources: Fr Survey, <http://minerals.usgs.gov/minerals/pubs/commodity>; Klee and Graede, 2004.)

^e B.Z. Holton (personal communication, 2011).

^f Biological N fixation on land and in the oceans totals ~300 Tg N/yr (see Figure 12.2).

^g Human enhancement in the atmosphere (Table 1.1) and rivers (Table 4.8).

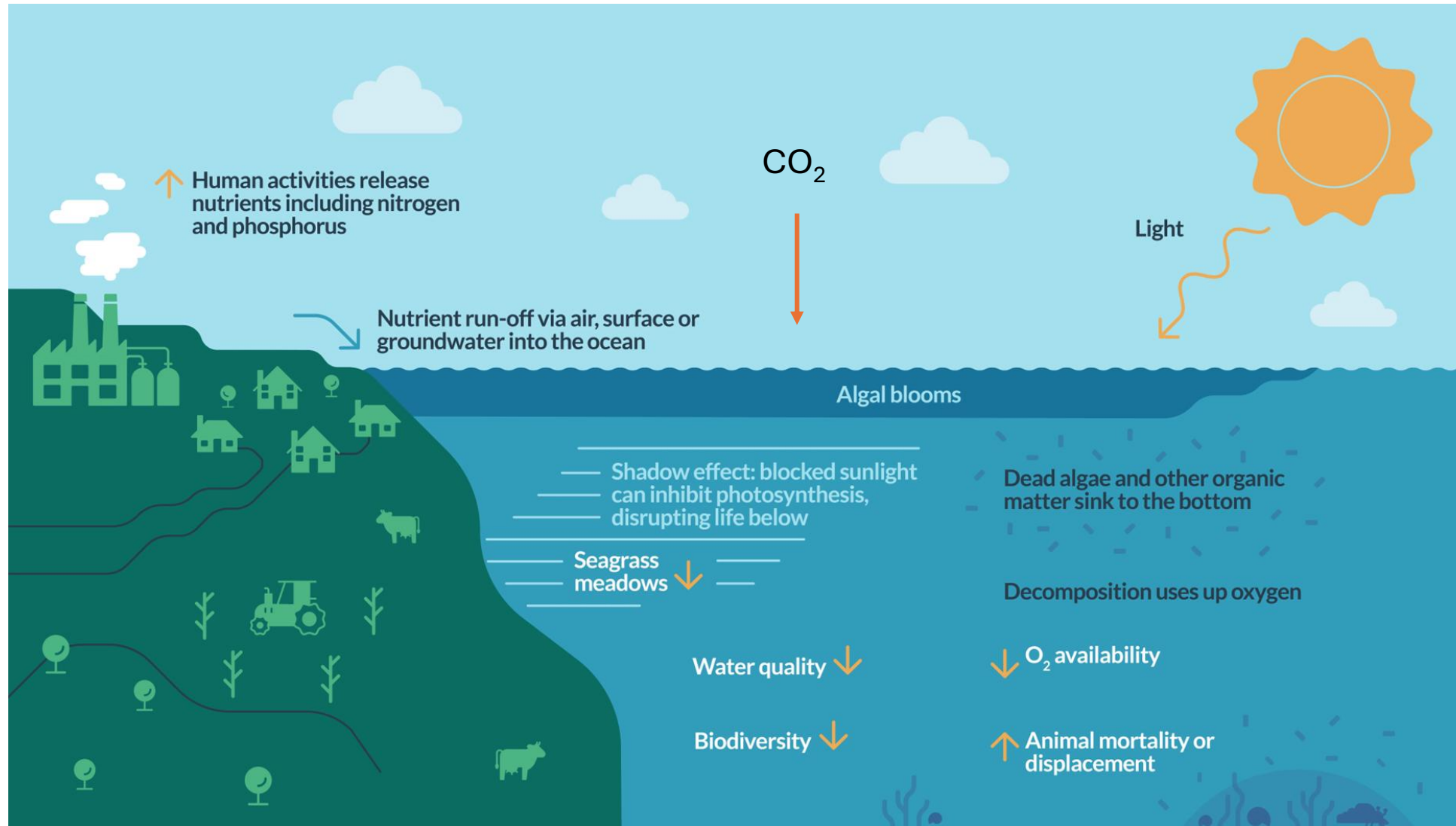


Net primary production versus nitrogen inputs to terrestrial, aquatic, and marine ecosystems

- It increases in direct response to added nitrogen up to inputs of about 10 g N m⁻² yr⁻¹ (100 kg/ha).
- Inputs in excess of that level are rarely found in natural ecosystems, but are seen in polluted environments and agricultural soils

Source: Biogeochemistry: An Analysis of Global Change

Eutrophication



It involves the over-enrichment of water bodies with nutrients like nitrogen and phosphorus, leading to excessive growth of primary producers (e.g., algae).

Impacts on aquatic ecosystems

- **Algal blooms:** block sunlight and inhibit photosynthesis
- **Oxygen depletion:** Aerobic decomposition of dead algae consumes large amounts of dissolved oxygen, creating low-oxygen or "**dead zones**"
- **Biodiversity loss:** Decrease in species diversity - sensitive species are outcompeted by more tolerant ones and the overall community composition shifts
- **Toxins:** Cyanobacteria (blue-green algae) produce toxins that are harmful to aquatic life, livestock, and humans
- **Changes in food webs:** Shifts in phytoplankton species and loss of submerged plants disrupt the natural food web, affecting species at higher trophic levels



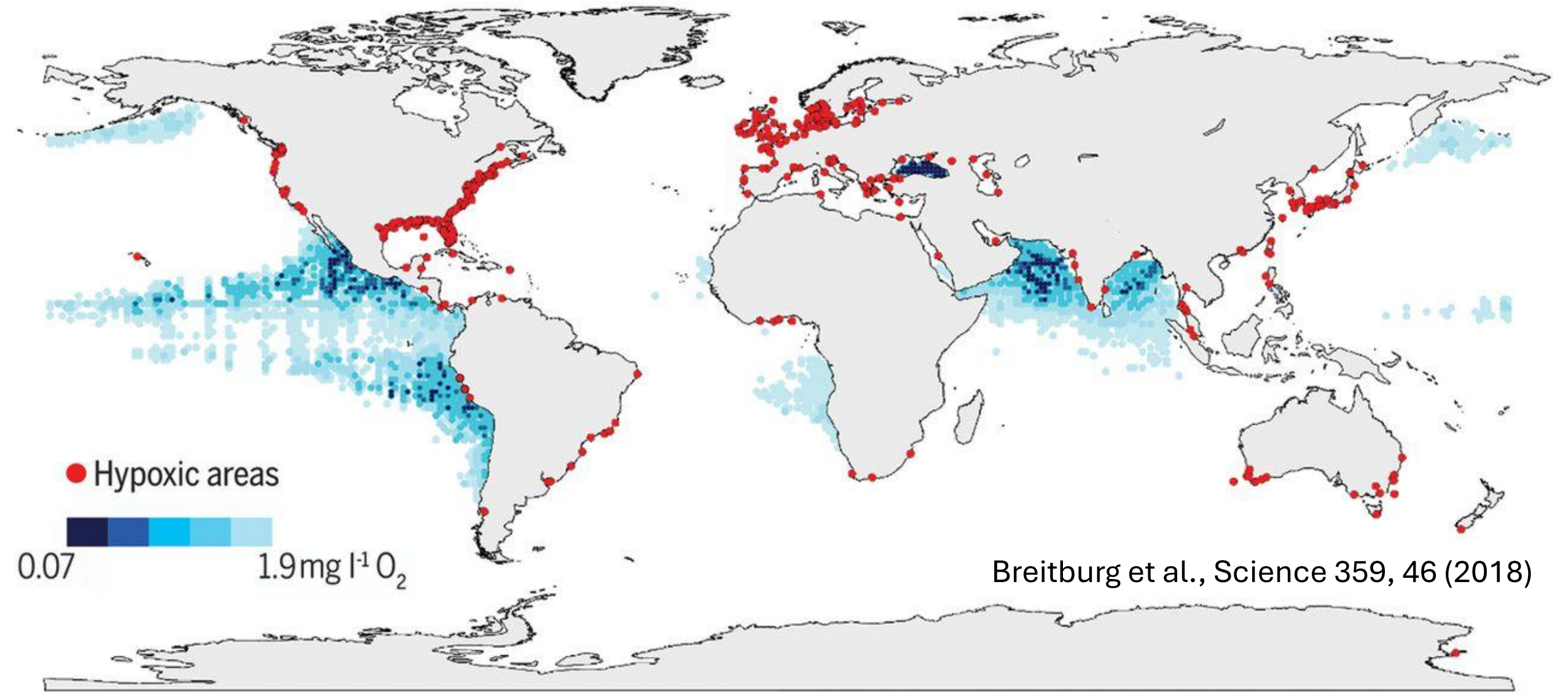
Most evident in terrestrial lakes and estuaries

Dal lake, with and w/o eutrophication

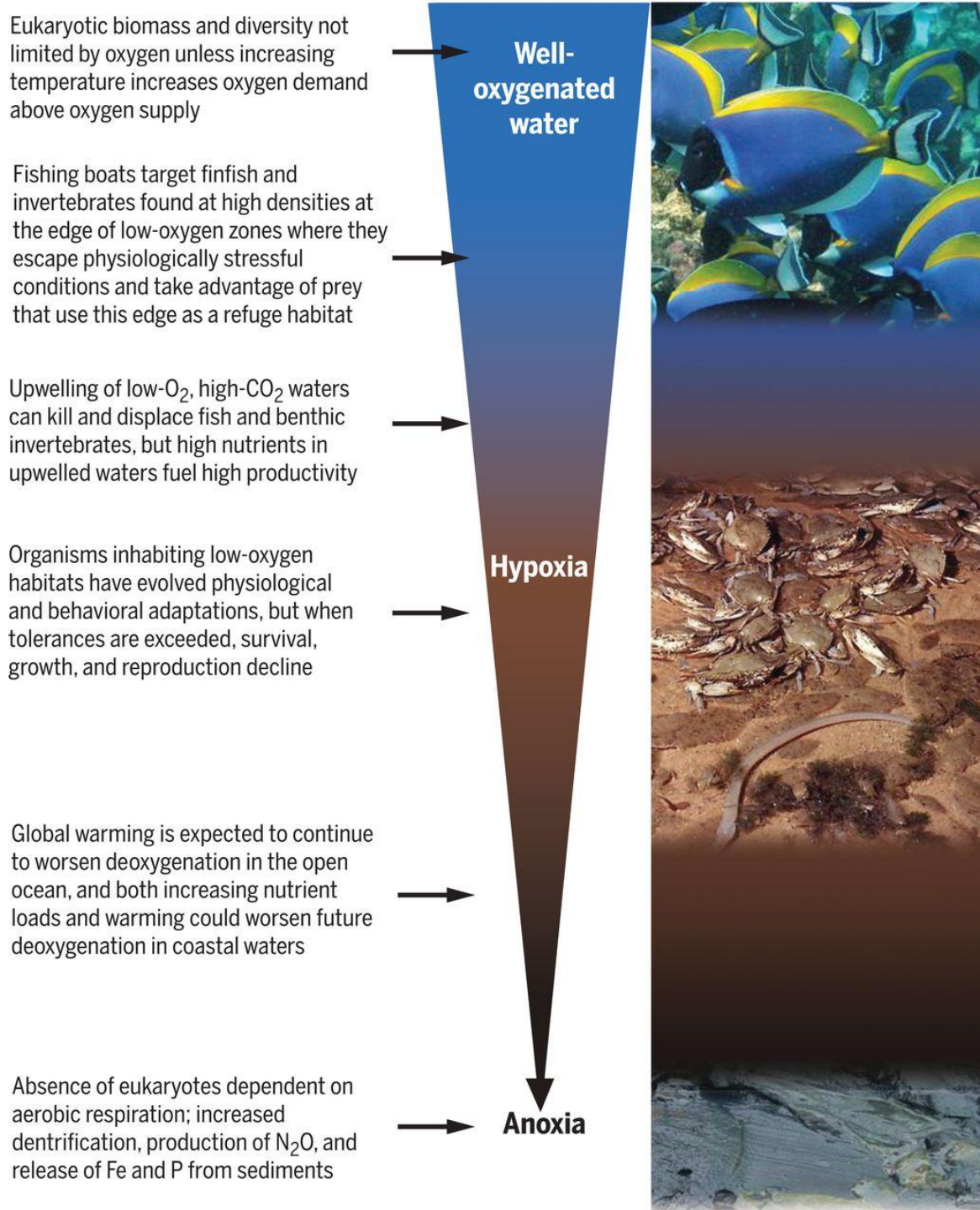


Pic credit:
Faisal Khan

Low and declining oxygen levels in the open ocean and coastal waters



Rapid increase in the aquatic dead zones across the world



Oxygen exerts a strong control over biological and biogeochemical processes in the open ocean and coastal waters.

(Top) Well-oxygenated coral reef with abundant fish and invertebrate assemblages.

(Middle) Low-oxygen event in Mobile Bay, United States, in which crabs and fish crowd into extreme shallows where oxygen levels are highest.

(Bottom) Anoxic mud devoid of macrofauna.

Breitburg et al., Science 359, 46 (2018)

Impacts on humans and the economy

- **Drinking water:** Algal blooms can contain toxins and create a foul odor and taste making water sources unfit for drinking.
- **Health risks:** Exposure to toxins from blooms can cause illness in humans and animals.
- **Economic losses:** The decline in fisheries and the unsuitability of water bodies for recreation activities – affects commercial and recreational fishing and tourism industries.
- **Water quality deterioration:** The accumulation of sediment and organic matter makes water bodies shallower over time

Deoxygenation management and policy strategies

Ecosystem-based mitigation to restore and protect the environment

Reduce greenhouse gas emissions to reduce deoxygenation due to climate change



Reduce anthropogenic nutrients reaching coastal waters to reduce eutrophication-driven deoxygenation

Develop aquaculture practices and limits to protect oxygen content of waters

Adaptation to restore and protect marine organisms and fisheries

Create marine protected areas and no-catch zones in well-oxygenated areas that can serve as refugia; protect populations when oxygen is low



Consider effects of low oxygen on production, non-fishing mortality, and fishing mortality in setting catch limits

Reduce fishing pressure on hypoxia-intolerant species. Utilize fishing gear that minimizes additional stress on oxygen-impacted fish stocks and ecosystems.

Implement and maintain monitoring and analysis programs

Monitoring, data analysis, and dissemination of results are critical to detect problems and determine the effectiveness of management and restoration efforts.



Strategies for deoxygenation management and policy-making

Breitburg et al., Science 359, 46 (2018)

Food production

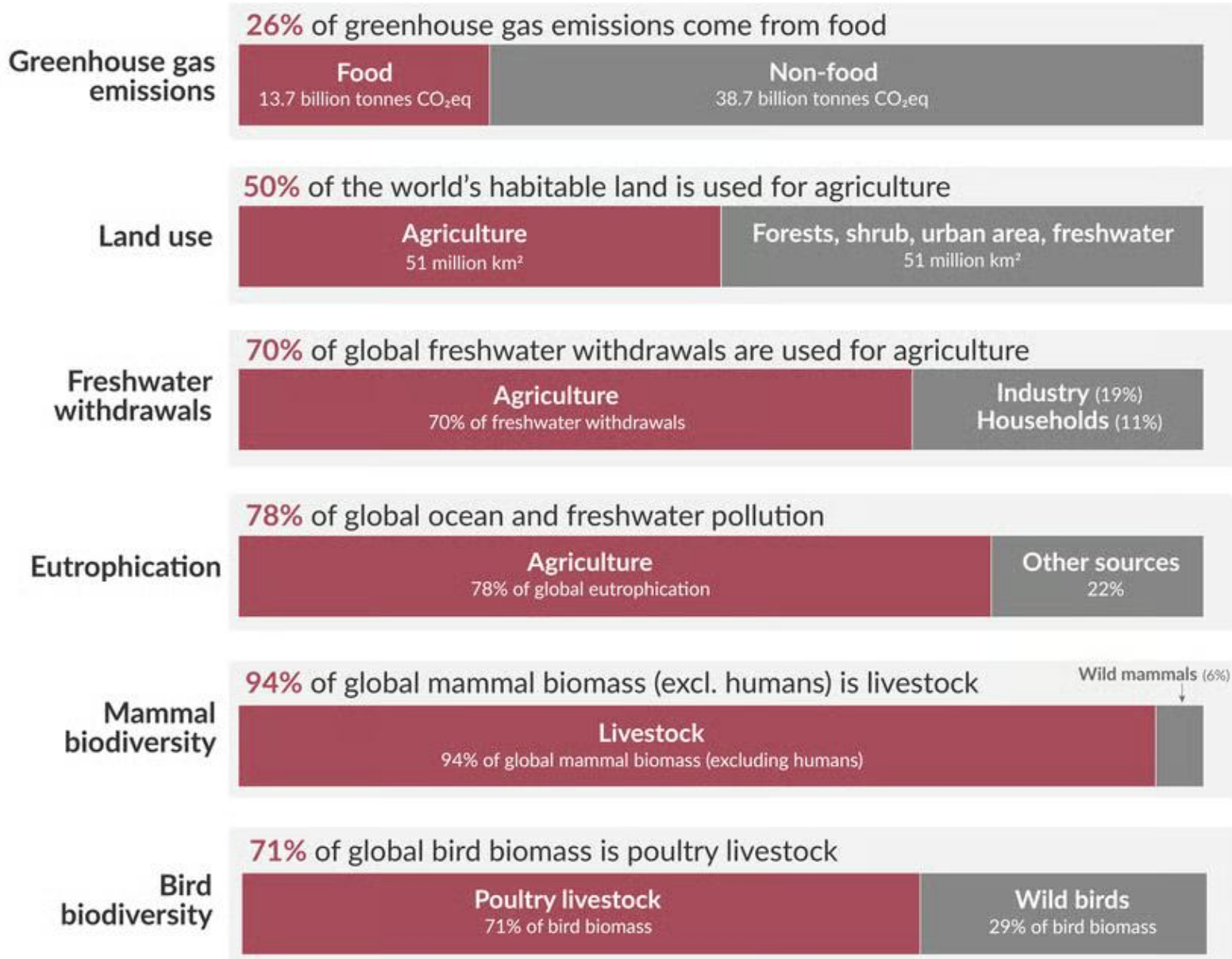
- Food production is influenced **directly by biogeochemistry via precipitation chemistry and changes in soil properties** induced by it
- Agriculture is sensitive to climate change.
- Impact of **rising CO₂ on crop productivity**

Human modifications

- **Large-scale agricultural changes:** green revolution (synthetic fertilizers, pesticides), high-yield crop varieties – loss of crop biodiversity
- **Industrial animal farming & aquaculture:** For dairy and meat, high-yield breeds – impacts environment
- **Selective breeding:** for developing new varieties of plants and animals with features like higher yield or greater size.
- **Genetic engineering:** Bt Corn resistant to pests

The environmental impacts of food and agriculture

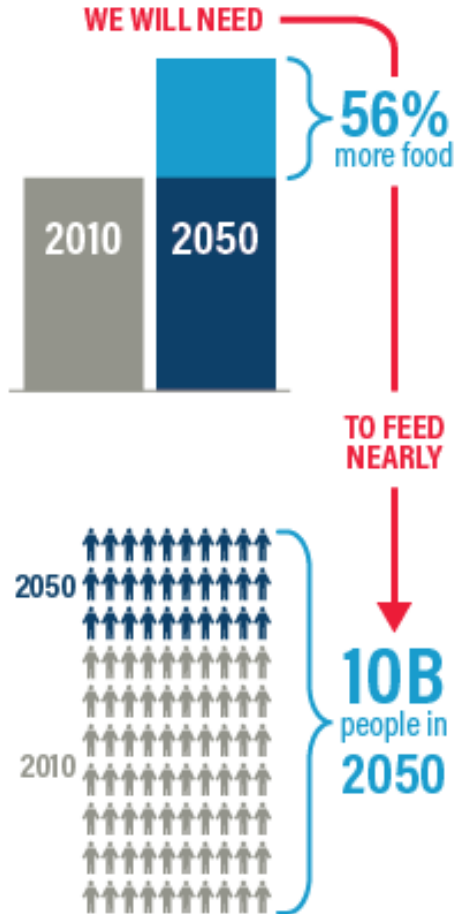
Our World
in Data



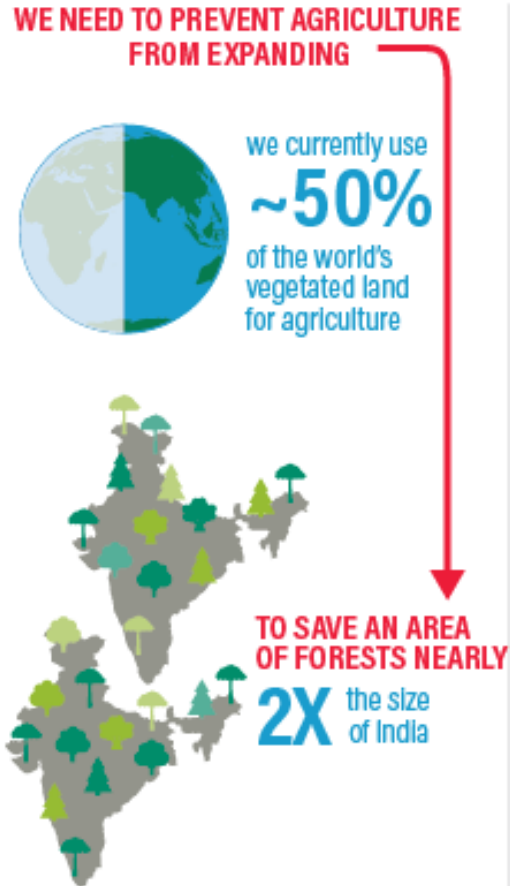
- Contributes about **26% of global greenhouse gas emissions**
- **Half of the world's habitable land** is used for agriculture.
- **70% of global freshwater withdrawals** are used for agriculture.
- **78% of global ocean and freshwater eutrophication** is caused by agriculture.
- 94% of non-human mammal biomass is livestock.
- 71% of bird biomass is poultry livestock.

CREATING A SUSTAINABLE FOOD FUTURE BY 2050

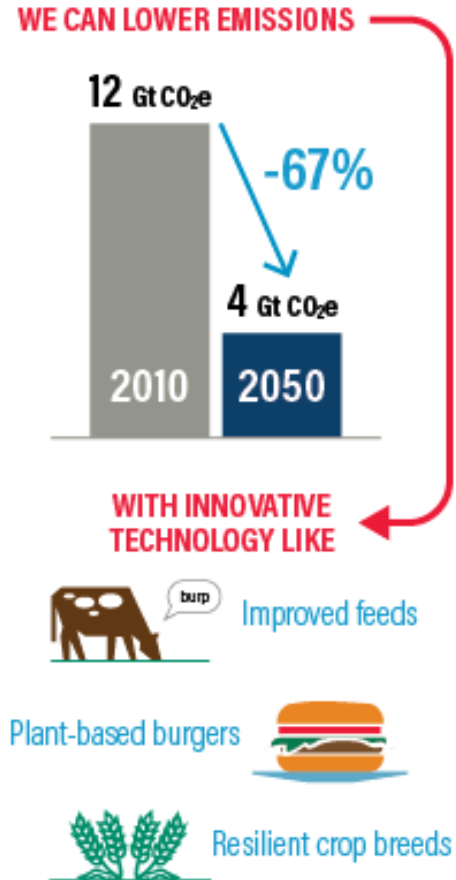
How do we feed
10 billion people...



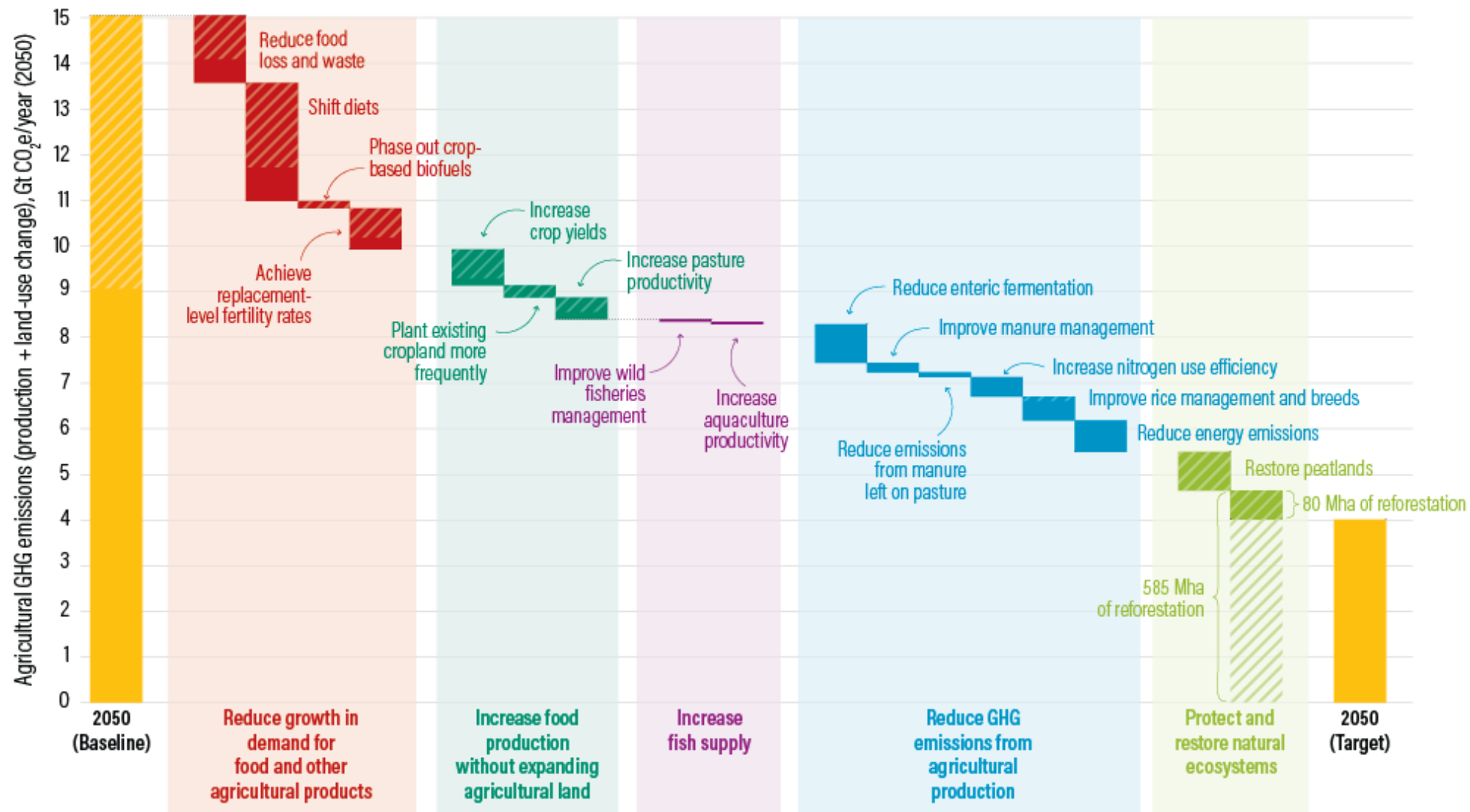
...without using
more land...



...while lowering
emissions?



A 5-Course Menu of Solutions Can Reduce Agricultural Emissions by More than 70%



Note: Solid areas represent agricultural production emissions. Hatched areas represent emissions from land-use change.

Source: GlobAgri-WRR model.